

PROMPT: I need you to complete the Python laboratory activity contained in the attached PDF named "Series1.pdf".

IMPORTANT:

- Read and study the ENTIRE PDF carefully before writing anything.
- The PDF contains 3 laboratory exercises.
- Follow the instructions, formulas, tables, numerical values, and terminology exactly as they appear in the PDF.
- Do NOT replace the given formulas with a different method unless absolutely necessary for Python implementation.
- Use Python with NumPy and Matplotlib where appropriate.
- The code must be beginner-friendly and easy for a college student to understand.
- Avoid unnecessarily complicated libraries or advanced programming techniques.
- Make sure the code actually runs without errors in a normal Python environment.
- Use comments in the code to explain important sections.
- The final graphs should closely reproduce the graphs/examples shown in the PDF in terms of information, labels, reference lines, and overall presentation.
- Do not fabricate numerical results. Calculate them using Python.
- Verify the numerical results before giving the final answer.

CLAUDE: A quick summary of what each does and one thing worth noting:

Exercise 1 — $(1 + 1/n)^n$

- Computes it for all 12 compounding frequencies (yearly → every nanosecond) and prints the table, matching the PDF's values exactly (2.000000, 2.250000, 2.441406, ...).
- Numerical note: naively computing $(1 + 1/n)^n$ in Python breaks down for huge n (nanosecond-level) because $1/n$ becomes smaller than floating-point precision can add to 1, so the result collapses to 1.0 instead of e . I fixed this using `math.exp(n * math.log1p(1/n))`, which stays accurate at every scale — verified all 12 rows converge correctly to 2.718282.

- Draws a two-panel chart: bar chart of values approaching e , plus a log-scale error chart, like the PDF's Figure 1.

Exercise 2 — $(a^h - 1)/h$ for $a = 2, e, 3$

- Reproduces the table exactly (0.7177, 0.6956, 0.6934, 0.6932... for $a=2$, etc.) down to $h = 1e-7$, with a $1e-6$ tolerance noted.
- Draws a grouped bar chart with dashed lines at each $\ln(a)$ limit, matching the PDF's Figure 2.

Exercise 3 — Taylor series for e^x at $x = 1$, up to $N = 10,000$ terms

- Uses an efficient running-term update ($\text{term} = \text{term} * x / n$) instead of recomputing factorials, which would overflow well before $N=10,000$.
- Prints partial sums at checkpoints, showing convergence to e (accurate to machine precision by $N=20$).
- Draws the histogram of partial sums plus a log-log "correct digits vs N " plot, matching the PDF's Figure 3.